Assignment 1 — Cryptocurrency Transaction Tracking and Anomaly Detection — Main Report Draft

COS30049 Assignment 1 — Project Management Plan and Prototype Design

Project: Cryptocurrency transaction tracking and anomaly detection

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1. Project Background and Introduction

Cryptocurrency transaction analysis has become an important area of cybersecurity research since examining transaction and address relationships can help researchers investigate unusual activity. However, research datasets contain substantial numbers of unknown labels, which makes it important to distinguish unusual patterns from confirmed malicious behaviour (Elmougy and Liu 2023).

This project proposes a research-oriented Bitcoin Address Activity Explorer. Its intended first user is a student researcher who needs to examine the historical activity of a supported Bitcoin address. Rather than manually navigating individual transaction records, the user will be able to explore an activity timeline, identify flagged transactions, and inspect the underlying transaction information and reasons for each flag. The project therefore aims to make historical Bitcoin activity easier to investigate and understand.

The intended output is an **unusual activity/anomalies indicator**, not a definitive judgement that an address or its owner has committed fraud. This distinction will guide the system's interface wording and the evaluation of its results.

This report covers Assignment 1, which establishes the project management plan, requirements and scope baseline, Work Breakdown Structure (WBS), schedule, risk and control plans, project closure approach, and a clickable front-end prototype with usability analysis. Assignment 2 will investigate and evaluate a machine learning method, while Assignment 3 will integrate the resulting pipeline into a React/FastAPI application, following

the unit's three-assignment sequence. No trained detection model or live service is claimed in this report.

1.1 Problem, user and objectives

As a research-oriented Bitcoin Address Activity Explorer, its explorer function exposes many transaction details, although the researcher still needs to distinguish which transactions deserve attention and why. The proposed tool combines a time view, transaction drilldown and an interpretable anomaly indication for an address selected from a documented historical sample. This in turn reduces the manual effort of scanning the sample, but it does not make identity, intent, ownership or criminality claims.

Objectives:

1. **Data sample.** Curate a snapshot of 20–30 Bitcoin addresses, each with at least 50 confirmed transactions over at least 90 days, plus 2–3 deliberately short-history addresses to exercise the insufficient-history state. Record every DR-01 field in a dataset manifest and a re-runnable retrieval script. Target: snapshot frozen by `Oct 25, 2026`; manifest reviewed by a second member.
2. **Baseline and ML methods.** Implement one rule-based baseline (robust z-score against the address's own trailing history) and two unsupervised ML methods, Isolation Forest (Liu, Ting and Zhou 2008) and Local Outlier Factor, all using only time-safe features. Target: all three score every transaction in the snapshot by `Oct 31, 2026`.
3. **Prototype journey.** Deliver three linked screens (address selection, daily timeline, transaction detail) and five feedback states (invalid, unsupported, loading, empty, insufficient history). Target: at least two participants complete all four walkthrough tasks, and every issue rated high severity is revised before submission.
4. **Qualified evaluation.** Evaluate the methods with the label-free plan in Section 2.3 (synthetic recall, rank stability, reviewer agreement). No accuracy, precision or recall against fraud labels is reported. Target: results table and limitations section in the Assignment 2 report.

Success measure for the whole project: AC-01 to AC-03 and FR-05 pass on every address in the snapshot in the Assignment 3 build, shown in the demo video.

1.2 Alternatives and decision gate

Approach	Potential value	Limit for this user journey	Current status
Curated Esplora histories (Blockstream n.d.)	Transaction IDs, timestamps, outputs and address history can support drilldown	Pagination, sample selection, provider limits, Bitcoin input/output	**Provisional direction; sample test required.**

	and transaction-level features.	interpretation, and scarce ground-truth anomaly labels.	
Bitcoin Heist address dataset (UCI Machine Learning Repository 2020)	Historical address/day features and ransomware-related labels.	Listed fields are aggregate address/day observations, not the individual transaction records required for this prototype; its “white” class is not known ransomware, not verified innocence.	Comparison source; cannot satisfy drilldown alone.
Elliptic++ (Elmougy and Liu 2023)	Bitcoin transaction/address graph research with temporal interactions.	Verify downloadable fields, identifiers and whether the supplied data can produce the specified user-facing transaction and time view.	Comparison source; not selected.

Approval gate D1: A Data/ML lead presents at least two candidate sources and one inspected sample with address, transaction ID, confirmed time, address-specific value, reproducible retrieval method, access/use terms and limitations. The four members then approve the primary source, fallback and scope. If no practical source supports the drilldown, the team must revise the promise and record a change before completing the report.

2. Project Requirement List and Description

2.1 Requirements hierarchy and project scope

The project requirements follow a structured hierarchy from the research question to user tasks, data requirements, project scope, system features, quality expectations and acceptance criteria. This approach supports requirements definition and traceability throughout the project lifecycle.

The selected unit of analysis is an individual transaction associated with one supported Bitcoin address. Transactions are grouped by day in the activity timeline. The system accepts

a Bitcoin address as input but does not claim to identify all addresses belonging to an entire wallet or individual.

2.2 Functional requirements

ID	Priority	Requirement	Phase and verification
FR-01	Must	A user can select a documented sample address; a typed address is accepted only if its history is in the supported dataset. The interface explains coverage.	A1 input and unsupported-address states; A3 input test.
FR-02	Must	A selected address displays a timeline of its observed confirmed transactions, with the date range and source/snapshot noted.	A1 timeline frame; A3 comparison with sample records.
FR-03	Must	Selecting a timeline period lists its individual transactions; selecting a transaction displays its ID, confirmed time, address-related amount/direction when derivable, and source link.	A1 linked detail frame; A3 record-level test.
FR-04	Must	Each transaction-level anomaly indication includes the contributing features, comparison	A1 annotated result; A2 analysis; A3 output test.

		basis and a statement that unusual does not mean illicit.	
FR-05	Must	Invalid, unsupported, loading, empty and insufficient-history states are distinguishable and offer a next action.	A1 clickable states and usability walkthrough; A3 tests.

2.3 Data requirements

ID	Priority	Requirement	Phase and verification
DR-01	Must	The team records source, collection date, sample selection, fields, missing values, provenance, use terms and retrieval instructions.	A1 data feasibility note; A2 reproducible dataset manifest.
DR-02	Must	Features for a transaction use only transactions confirmed before it. Models are fitted on the earlier part of the snapshot and evaluated on the later part. **Pass:** (1) a unit test recomputes F1–F4 for 20 random transactions using only earlier records and gets identical	A1 plan; A2 code review by a member other than the author.

		values; (2) no transaction ID appears in both the fit and evaluation periods; (3) the Assignment 2 report states how sample addresses were chosen and any resulting bias.	

Anomaly definition

A transaction is flagged when it is unusual compared with the same address's own earlier activity. The amount and direction rule below is confirmed against the first real sample at decision gate D1.

Address-related amount and direction: received = sum of outputs paying the address; sent = sum of inputs spent from the address; net = received minus sent. Positive net is shown as “in”, negative as “out”.

ID	Feature	Definition (only transactions confirmed before the one scored)	Why
F1	Size	Absolute net amount in BTC, log-scaled	Unusually large or small transfers
F2	Burst	Number of the address's transactions in the previous 7 days	Sudden bursts of activity
F3	Gap	Hours since the address's previous transaction	Dormant address waking up

F4	Relative size	Absolute net amount divided by the median absolute net amount of the previous 30 days	Size relative to the address's own habit

Minimum history: at least 10 earlier transactions spanning at least 14 days. Below that the transaction is not scored and the interface shows the insufficient-history state.

Baseline method: for each feature, robust z = (value − median of previous 30 days) / (1.4826 × median absolute deviation of previous 30 days). A transaction is flagged when any feature has an absolute robust z of 3.5 or more. The 3.5 starting threshold is a team choice and is tuned in Assignment 2.

ML methods: Isolation Forest (Liu, Ting and Zhou 2008) and Local Outlier Factor on F1–F4, fitted on the earlier part of the snapshot only. The flag threshold starts at the top 2% of scores and is tuned with the synthetic-injection results below.

Detail panel (FR-04, AC-03): shows each feature with an absolute robust z of 2 or more against the address's usual range; the comparison basis (previous 30 days, number of transactions); and the fixed line “Unusual activity requires investigation; it does not indicate wrongdoing.”

Assignment 2 evaluation plan without labels

The methods are compared on four checks that need no fraud labels, all run on the later part of the snapshot only.

Check	Method	Measure	Target
Synthetic injection	Insert 50 synthetic transactions of each pattern: value spike (10× the address median), burst (10 transactions in one day), dormant wake-up (after 60+ idle days). They	Share of injected transactions flagged	70% or more for at least two of the three patterns

	are marked as synthetic.		
Stability	Re-run each ML method with 5 random seeds and 5 bootstrap resamples	Jaccard overlap of the top 50 flagged transactions between runs	0.7 or more
Method agreement	Compare flagged sets of the baseline, Isolation Forest and Local Outlier Factor	Pairwise Jaccard overlap; transactions flagged by only one method listed	Reported, no target
Reviewer check	Two members independently rate 20 top-flagged and 20 random unflagged transactions without seeing which is which	Share of flagged rated “worth a look”; agreement between reviewers (Cohen's kappa)	Reported as reviewer judgement, never as ground truth

Sanity rule: no single address should have more than 20% of its transactions flagged; if one does, the threshold or features are revisited before results are used.

Selection rule: choose the method with the highest synthetic recall among those meeting the stability target. If two methods are within 5 percentage points, prefer the one whose flags are easier to explain in the detail panel.

2.4 Non-functional requirements

ID	Priority	Requirement	Phase and verification
NFR-01	Must	Charts and the anomaly view state units, source and uncertainty, and never imply guilt.	A1 heuristic review and walkthrough; A3 usability test.

		Pass: (1) every chart has a title, axis labels with units (BTC, count, date) and the snapshot date; (2) the detail panel shows the fixed limitation line on every flagged transaction; (3) a wording check finds none of: fraud, criminal, illicit, guilty, suspicious; (4) at least two walkthrough participants correctly say a flag means “needs investigation”, not “wrongdoing”.	
NFR-02	Should	The sample and feature process can be reproduced by another team member using documented versions and procedures.	A2 reproduction exercise.
NFR-03	Should	The interface supports keyboard navigation, visible focus, readable contrast and recovery from mistakes.	A1 design checklist; A3 accessibility inspection.

Acceptance criteria

AC-01 — supported sample: Given a researcher is on the start screen, when they select a documented sample address, then the timeline shows the observed date range, labelled transaction counts, and a clear path to the flagged transaction list.

AC-02 — unsupported input: Given an address outside the curated sample, when submitted, then the interface explains the coverage limit and offers supported examples without inventing a result.

AC-03 — anomaly interpretation: Given a transaction is flagged, when its details open, then the researcher can see the actual contextual features, comparison basis and a “requires investigation” limitation.

2.5 Association with WBS and Gantt chart

Requirement	WBS work packages	Gantt phase and due date	Verified by
FR-01 Supported-address selection	1.1.6 prototype; 1.2.1 sample defines coverage; 1.4.2 back-end validation; 1.4.3 input form	A1 (11 Oct), A3 (6 Dec)	AC-02; 1.4.6 test log
FR-02 Daily timeline	1.1.6; 1.4.3; 1.4.4 charts	A1, A3	AC-01
FR-03 Transaction drilldown	1.1.2 source check (D1); 1.2.1; 1.4.2; 1.4.3	A1, A2 (8 Nov), A3	AC-01; record-level comparison with the snapshot
FR-04 Anomaly indication	1.1.6 annotated result; 1.2.4 models; 1.2.5 evaluation; 1.4.2 scoring endpoint	A1, A2, A3	AC-03; Assignment 2 evaluation checks
FR-05 Feedback states	1.1.6; 1.4.2; 1.4.3; 1.4.6	A1, A3	Walkthrough task 4; 1.4.6 test log
DR-01 Source record and manifest	1.1.2; 1.2.1; 1.2.2; 1.2.6 README	A1, A2	Manifest reviewed by a second member
DR-02 Time-safe features	1.2.2; 1.2.4; 1.2.5; 1.3.5 review	A2	DR-02 pass conditions (1)–(3)
NFR-01 Units and uncertainty wording	1.1.6; 1.4.4; 1.4.6	A1, A3	NFR-01 pass conditions (1)–(4)
NFR-02 Reproducibility	1.2.6; 1.3.5	A2	Second member re-runs the README

			from a clean environment
NFR-03 Accessibility	1.1.6; 1.4.3; 1.4.6	A1, A3	Keyboard-only run of the journey; contrast check

2.6 Planned Assignment 3 architecture and API

Esplora is called once, offline, to build the snapshot. A snapshot script (paged and rate-limited) feeds the feature and scoring pipeline (F1–F4, baseline, Isolation Forest, Local Outlier Factor), which writes the snapshot, scores and manifest to a local store once. In the demo, the React front end calls the FastAPI back end with GET and POST requests, and FastAPI only reads the local store. No live blockchain calls are made from the application.

Method	Path	Returns	Serves
GET	/api/addresses	Supported sample addresses, date range, snapshot date	FR-01, AC-01
POST	/api/lookup	For a typed address: supported, unsupported or invalid, with a message	FR-01, FR-05, AC-02
GET	/api/addresses/{address}/timeline	Daily transaction counts, flagged days, source and snapshot date	FR-02
GET	/api/addresses/{address}/transactions?date=	Transactions for one day, with flag status	FR-03
GET	/api/transactions/{txid}?address=	ID, confirmed time, amount and direction, features, comparison	FR-03, FR-04, AC-03

		basis, limitation line, source link	

Error states (FR-05): 400 invalid format; 404 address outside the sample; 422 insufficient history (no score). The front end maps each to its feedback state, and a loading state covers every request.

3. Team and communication plan

Role	Accountable outputs	Reviewer
Leader / report integrator — Cam Dan Bui	Problem statement, decision log, rubric, report integration, meeting follow-up and submission	PM lead
Data and ML lead — Cao Tri Tran	Source audit, feature/data contract, baseline and evaluation plan, data risks	UX lead
PM and schedule lead — Le Quang Anh Hoang	Scope, WBS/dictionary, estimates, Gantt/critical path, risk/change register	Leader
UX and web lead — Minh Khang Nguyen	User flow, clickable prototype, heuristics, usability review and revisions	Data lead

Communication rule: The team holds one weekly decision meeting with a written agenda and genuine minutes (date, attendees, discussion, decision, action, owner and due date), following the Week 2 communication guidance. Owners update the task board before the meeting. Reviewers comment on evidence within one working day where possible; a blocked task is raised in the shared workspace the same day. Material scope or data changes go into the change log and require recorded agreement by the four members, with unresolved academic requirements referred to the tutor. Minutes are verified by attendees before they are attached.

Weekly records: Meeting Minutes 1, Meeting Minutes 2, Meeting Minutes 3.

3.1 Stakeholders

Stakeholder	Interest	Influence	How engaged

Project team (four members)	Deliver A1–A3 on time and share work fairly	Decides scope, schedule and changes	Weekly meeting, minutes, contribution form
Tutor and unit convenor	Work meets briefs, rubric and academic integrity	Final academic acceptance; approves topic and group size	Canvas submissions; questions by message, logged in the decision record
Student researcher (target user)	Find and understand unusual activity for one address	Shapes requirements; not a real client	Represented by walkthrough participants
Walkthrough participants	Clear tasks, short sessions	Their findings drive prototype revisions	Four-task walkthrough; findings recorded in Section 10
Data provider (Blockstream Esplora)	Fair API use within its terms	Can limit or block retrieval	Rate-limited, one-off snapshot; terms recorded in the manifest
People behind sample addresses	Not to be identified or accused	None directly, but carry the ethical risk	No identity claims; fixed limitation line; excluded scope items

4. Scope statement and baseline

Scope of the project in this section is defined as what the project will and will not produce. Items are suggested for team approval at decision; no approval is recorded here.

4.1 Scope statement and deliverables

Project scope. The project builds a research-focused Bitcoin Address Activity Explorer: one supported Bitcoin address at a time from a curated sample, an activity timeline aggregated

per day, drilldown to individual confirmed transactions, and a transparent transaction-level indication of unusual activity that shows its contributing features and a limitation statement. The work is delivered across three assignments.

Success boundary. The final system can explain why the observed transactions in its selected sample stand out; it cannot independently establish wrongdoing.

Assignment 1 (Project Management & Prototype): Scope baseline, WBS & dictionary, Gantt chart/critical path, risk, quality, monitoring, change control, and closure plans; usability explanation; meeting minutes; contribution form.

Assignment 2 (ML Pipeline & Data): Versioned data sample & processed dataset, time-safe feature pipeline, 2+ ML models (justified & evaluated), commented code, README (setup, training, prediction), 3,000-word report, meeting minutes, contribution form.

Assignment 3 (Web App & Deployment): Web app integrating the model (validated input form, address timeline, transaction drilldown, 2+ chart types, error handling, backend with 2+ HTTP methods), README, 8-page report with API docs, 7-min responsive demo video (desktop/tablet/mobile), meeting minutes.

4.2 Minimum viable demonstration

Proposed for approval, the minimum viable demonstration is one complete journey through the Explorer:

1. The researcher selects a documented sample address; an address outside the sample receives an explanation, not a result.
2. The daily timeline shows the observed date range and the source or snapshot used.
3. Selecting a period lists its transactions; selecting one shows its ID, confirmed time, address-related amount and direction, and a source link.
4. The anomaly indication shows its contributing features, its comparison basis and the statement that unusual does not mean illicit.
5. Invalid, empty, loading and insufficient-history states are distinguishable and offer a next action.

The demonstration is met when AC-01 to AC-03 and FR-05 (Section 2) pass on documented sample records. Everything classed as optional below is considered only after this journey works end to end.

4.3 Scope exhibit: must-have, optional and excluded

Class	Item	Phase	Requirement	WBS	Reason

Must-have	Reproducible data sample (metadata, terms, retrieval steps) with processed dataset and README	A2	DR-01, NFR-02	1.2.1, 1.2.2, 1.2.6	Required for A2; ensures repeatability .
Must-have	2+ justified ML methods (e.g., Isolation Forest) evaluated honestly without unlabelled claims	A2	FR-04	1.2.4, 1.2.5	Meets A2 criteria; data lacks ground-truth labels
Must-have	Address daily activity timeline & detailed transaction list/view	A1 prototype, A3	FR-02, FR-03	1.1.6, 1.4.3	Core MVD requirement
Must-have	Front-end and back-end validation with proper states (invalid, loading, empty, error)	A1 prototype, A3	FR-05	1.1.6, 1.4.2, 1.4.3, 1.4.6	A3 requirement and usability standard
Must-have	React FE + FastAPI BE (2+ HTTP methods), 2+ charts, responsive layout	A3	NFR-01	1.4.1 to 1.4.4	Required by A3 brief and demo video.

	(desktop/tablet/mobile)				
Must-have	All required deliverables (A1–A3 reports, prototype, code, READMEs, API docs, video, minutes, contribution forms)	A1 to A3	–	1.1.8, 1.2.6, 1.2.7, 1.3.1, 1.3.7, 1.4.7	Marked deliverables.
Optional	3rd chart type and advanced interactive features (zoom, filtering, tooltips)	A3	NFR-01	1.4.4	Bonus A3 marks; not needed for the minimum viable demonstration.
Optional	Additional ML model experiments beyond the core two	A2	FR-04	1.2.4	Bonus A2 marks; adds tuning and report effort.
Optional	Merging or comparing extra public datasets	A2	DR-01	1.2.1, 1.2.2	Bonus A2 marks; adds merge effort and licence checks.
Excluded	Non-Bitcoin chains and	–	FR-01	–	Preserves scope;

	addresses outside the curated sample				unsupported inputs return explanation
Excluded	Identity de-anonymization, wallet tracking, or private wallet monitoring	–	–	–	Ethics and data limitations

WBS codes refer to the WBS and dictionary in Section 5. Requirement IDs refer to Section 2.

4.4 Assumptions and constraints

Assumptions

- The tutor-approved topic remains acceptable across the three assignments.
- Members have access to Confluence and the development tools, and the tools used are free or open source.
- The due dates and requirements quoted in 4.1 are those on Canvas; any difference is confirmed and recorded.
- Role owners and estimates are provisional until the team approves its name mapping.

Constraints

- The unit runs across 12 teaching weeks: Assignment 1 is planning and prototyping, Assignment 2 is the data and model, and Assignment 3 is full-stack integration (Li n.d.a, slides 5 and 8).
- Due dates are 11 Oct, 8 Nov and 6 Dec 2026 at 23:59. Late work loses 10% of the available mark per calendar day, and work more than seven days late scores zero (Swinburne University of Technology n.d.a, n.d.b, n.d.c).
- A team of four works part-time alongside other units, with no financial budget.
- Format limits: Assignment 1 report requires a minimum of 800 words, the Assignment 2 report is up to 3000 words, the Assignment 3 report is up to 8 pages, and the demo video is up to 7 minutes and must be accessible to markers.

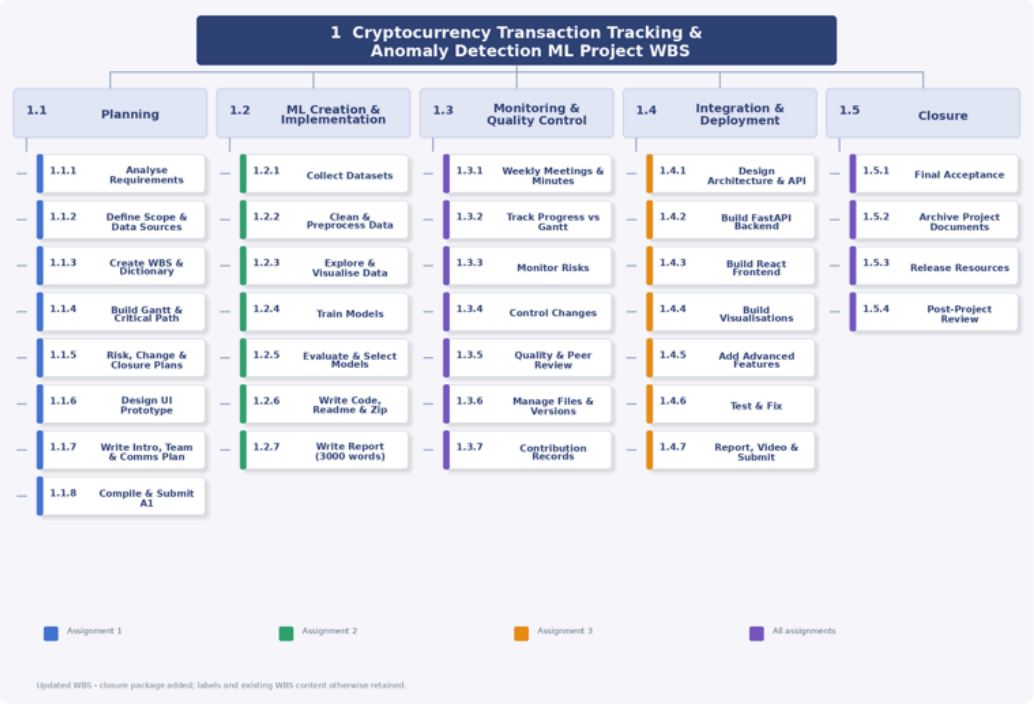
4.5 Scope validation

Validation. Deliverables are verified and formally accepted, not assumed complete (Li n.d.c, slides 14–15):

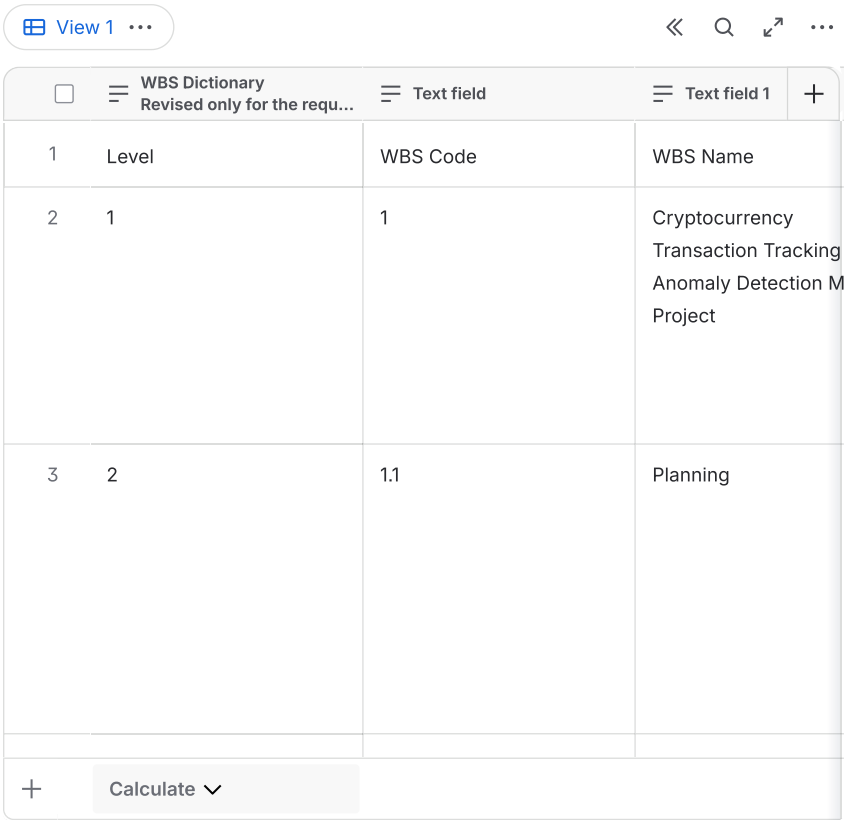
- Each deliverable is checked against its requirement, acceptance criteria and the unit rubric by a reviewer other than its author; the reviewer and the evidence are linked on the task board.
- The prototype and, later, the application are walked through against AC-01 to AC-03 and FR-05.
- The four members record acceptance at a weekly meeting. The tutor remains the academic acceptance authority.

5. Work Breakdown Structure and dictionary

WBS hierarchy (deliverable-oriented):



WBS Dictionary



6. Time management, Gantt and critical path

The **provisional 12-week Gantt** places A1 planning and prototype in Weeks 1–4, A2 data/ML in Weeks 4–8, and A3 web integration and tests in Weeks 8–12. A2 starts after the A1 submission (WBS 1.1.8); A3 work starts once the A2 method is selected (1.2.5), so it overlaps the A2 code packaging and report. A1 is due on 11 October 2026 at 23:59; the time zone is to be confirmed on Canvas. Figure 1 presents WBS codes and weekly bars, and Figure 2 shows the dependencies and the critical path, following the Week 2 scheduling material (Li n.d.c, slides 20–26; Li n.d.d, slides 12–14) and standard critical path practice (Project Management Institute 2011). Durations are planning estimates in calendar days, matching the 84-day unit. Some draft work is already complete, so the remaining work is re-estimated against member availability.

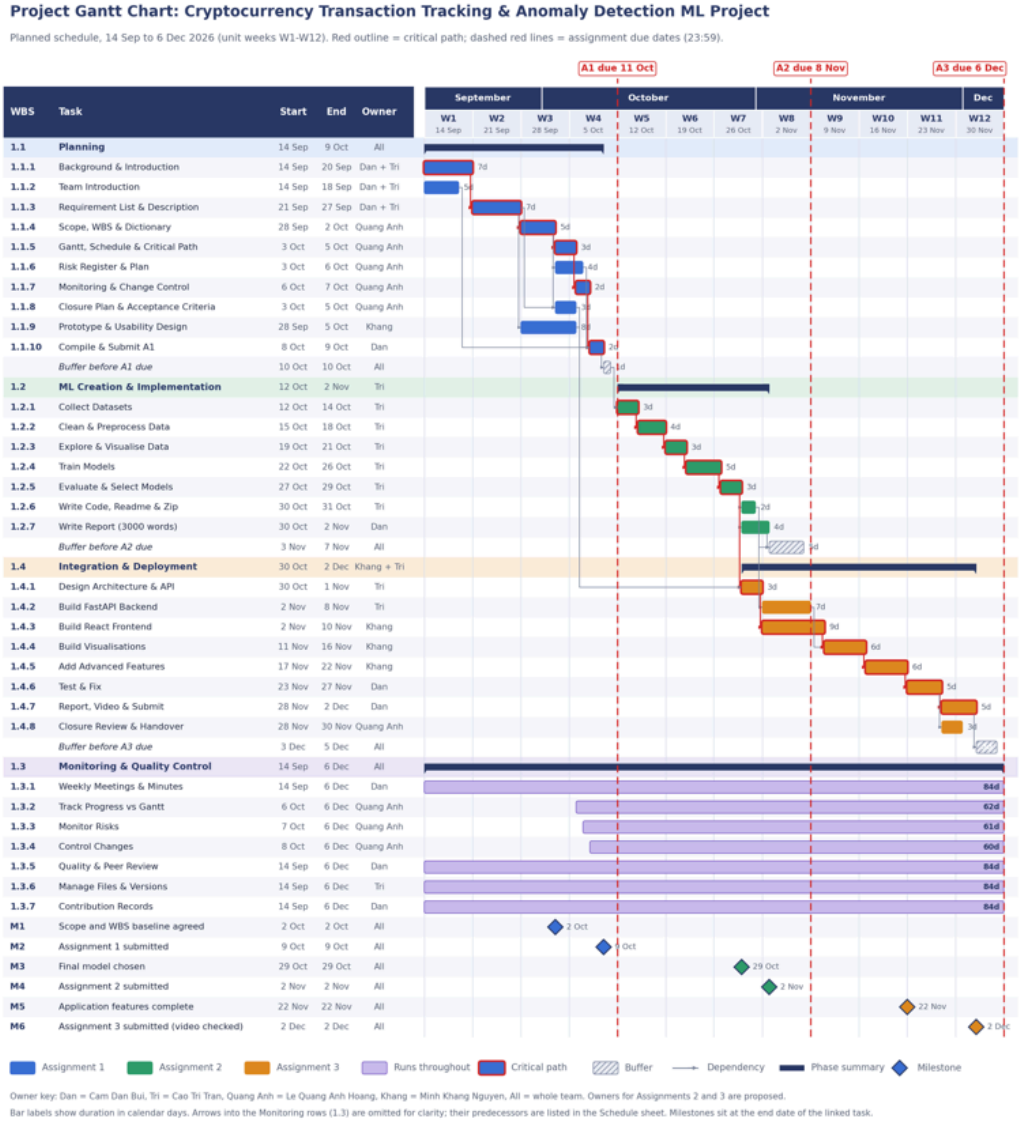


Figure 1. Gantt chart for Assignments 1–3 across 12 weeks

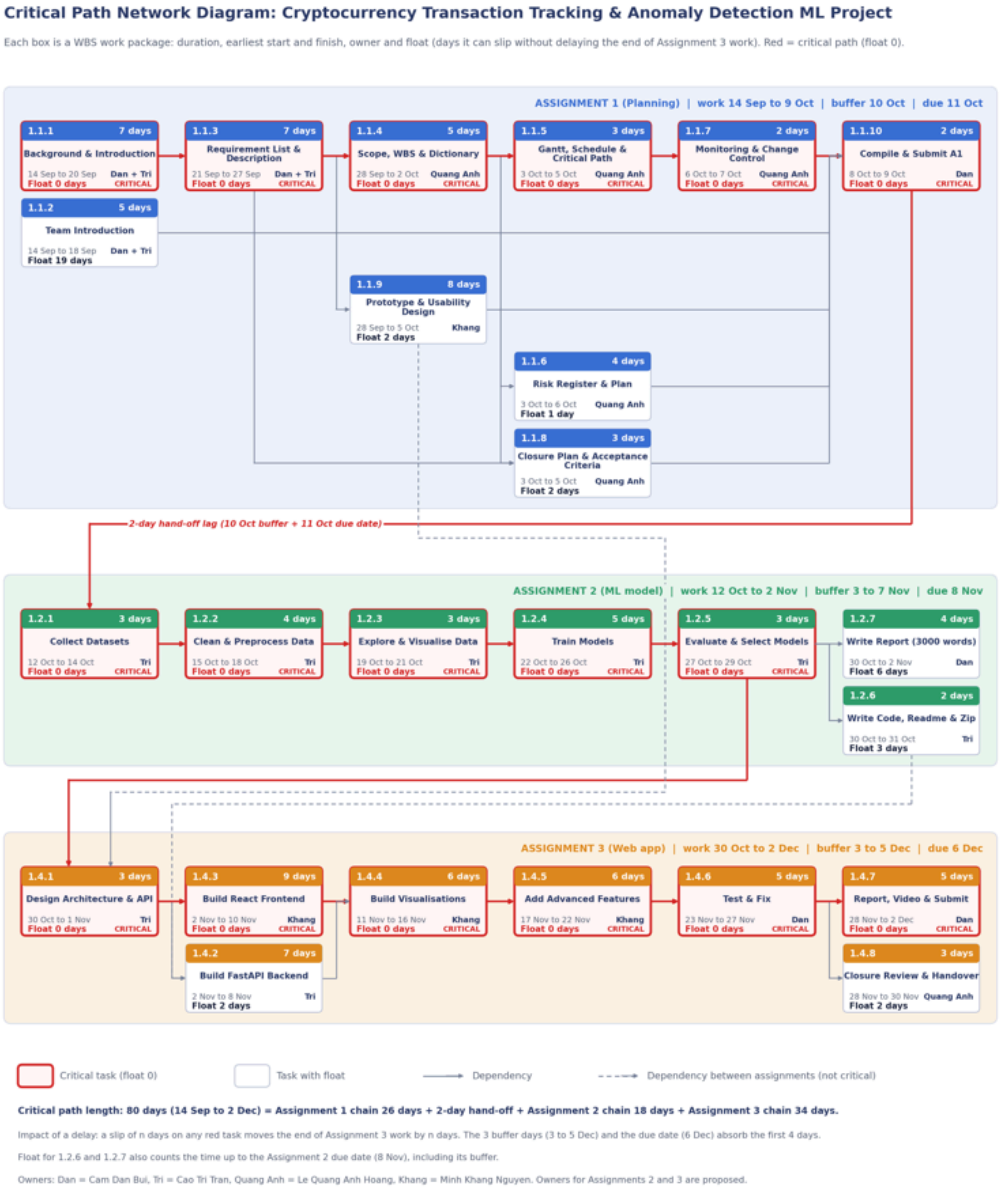
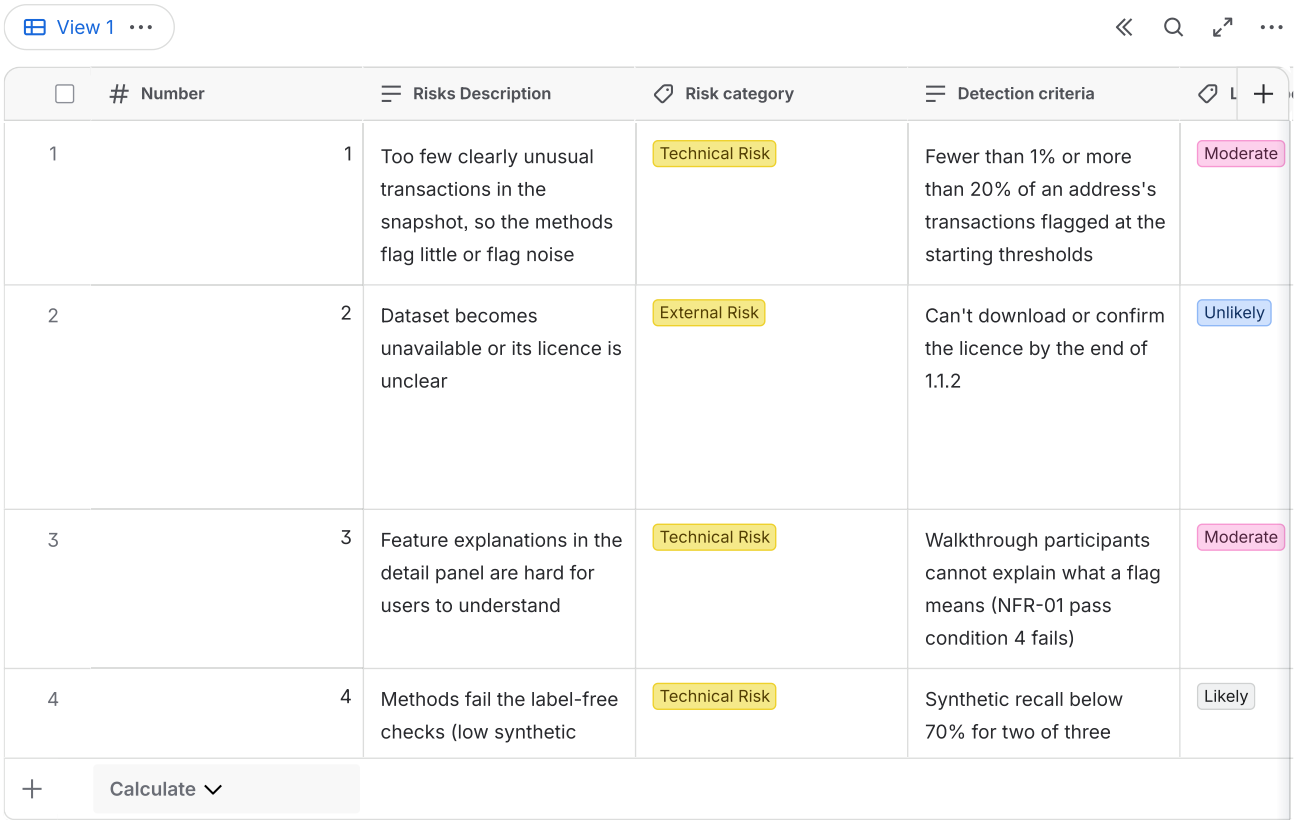


Figure 2. Critical path

7. Risk and quality management



View 1

	Number	Risks Description	Risk category	Detection criteria	L
1	1	Too few clearly unusual transactions in the snapshot, so the methods flag little or flag noise	Technical Risk	Fewer than 1% or more than 20% of an address's transactions flagged at the starting thresholds	Moderate
2	2	Dataset becomes unavailable or its licence is unclear	External Risk	Can't download or confirm the licence by the end of 1.1.2	Unlikely
3	3	Feature explanations in the detail panel are hard for users to understand	Technical Risk	Walkthrough participants cannot explain what a flag means (NFR-01 pass condition 4 fails)	Moderate
4	4	Methods fail the label-free checks (low synthetic	Technical Risk	Synthetic recall below 70% for two of three	Likely

Calculate

Quality plan: Section authors verify claims against primary sources and add Harvard citations as they draft. A separate reviewer checks that requirements, WBS, dictionary and Gantt use the same scope and codes; the data lead checks feature definitions and leakage; the UX lead runs a scenario walkthrough; the leader conducts a final rubric/grammar/export review. In A2, a different member reproduces at least one sample and model result from the documented steps. Do not mark work complete merely because text or a graphic exists: link the accepted evidence and reviewer in the task board.

8. Monitoring and change control

Monitoring and controlling means tracking, reviewing and regulating progress so that problems are found early and the plan changes deliberately, not by accident . The team compares actual progress with four baselines: the scope baseline (Section 4 and the WBS dictionary in Section 5), the schedule baseline (the Gantt chart and critical path in Section 6), the risk register (Section 7) and the acceptance criteria (Section 9). The team has four members and no formal hierarchy, so the work is shared: every package owner reports on their own packages, and all members attend the weekly meeting, reviewing each other's work.

8.1 Weekly monitoring cycle

Step	When	Who	What happens	Output
1. Update	Before each weekly meeting	Each package owner	Update % complete, effort used and	Updated task board

			blockers on the task board.	
2. Review	Weekly meeting on Discord (W1 to W12)	All four members	Compare planned and completed packages with the Gantt chart, starting with critical-path packages. Review blockers, risk triggers, open change requests and review findings.	Minutes with decisions, actions, owners and due dates
3. Record	Within one working day of the meeting	Project manager	Update progress on the Gantt chart, calculate the indicators and update the risk register status.	Updated Gantt chart, indicators and registers
4. Act	At the meeting or within the week	Owner of the affected package	Apply the response in Table 8.3. Raise a change request if a baseline must change.	Corrective action or change request

The minutes of each meeting are checked by the attendees before they are filed, so that the record shows what was actually discussed and decided.

8.2 Responding to variance

Situation	Response	Decided by
A package is late but within its float	Re-plan inside the week and update the Gantt chart.	Package owner

A critical-path package is late	Reassign work, use buffer days if needed, record the decision in the minutes and update the critical path.	Weekly meeting
More than half the buffer is used, or SPI is below 0.8	Defer or remove optional items through a change request, and tell the tutor early if a due date is at risk.	Change request, approved by the team
A review fails	The owner fixes the work and the reviewer rechecks it within two working days.	Reviewer
A risk trigger fires	Start the contingency plan in the risk register and update the risk status.	Risk owner

8.3 Quality control

Quality is controlled through the same weekly cycle with every deliverable is reviewed by someone other than its author against the unit rubric and its acceptance criteria. The evidence and reviewer are linked on the task board, and a package is not marked complete because a document or graphic merely exists. Code and README are checked by a clean installation by another member, and the prototype is checked by a walkthrough with at least two people who did not design it.

8.5 Risk monitoring

The risk register in Section 7 is reviewed at every weekly meeting. Each risk has a status (open, mitigating, triggered or closed), and likelihood and impact are re-scored after the data source decision and at the start of Assignments 2 and 3. Four risks need particular attention because they can be missed until late:

Risk	Where it is checked	Trigger	WBS
Predictions misread as proof that someone committed fraud	Every results screen and every report section, before the demo is recorded	Wording that suggests guilt, or a result without the “unusual does not mean illicit” statement	1.4.3, 1.4.7

Weak or noisy labels lead to misleading model training or evaluation	When the sample is audited, and before the evaluation is written	Labels cannot be traced to a source, or a result claims fraud precision or recall without valid labels	1.2.1, 1.2.5
External or live API limits prevent reliable requests or demo execution	When the snapshot is collected, and before the video is recorded	Failed or rate-limited requests, or the application needing a live call to run	1.2.1, 1.4.2, 1.4.7
Data rights or privacy requirements are unclear, or the data contains information that should not be exposed	When the source is chosen, and before any record is shown in the interface	Licence or use terms not recorded, or a field that could identify a person	1.1.2, 1.2.1

8.6 Change control

Change control makes sure that changes to the baselines are reviewed, approved and recorded, so that the scope, schedule and risks may stay consistent. The lecture describes a Change Control Board and a change log, the four members act as the change control group, and the change log is the record.

What needs a change request. A change request is needed to change a must-have item or add an optional item from the scope exhibit to move a package's end date by more than its float; to add or remove a package; to change a package owner or its acceptance criteria; or to use more than half of an assignment buffer.

Approval rule (proposed; to be confirmed by the group). A change is approved when at least three of the four members agree, including the owner of the affected package, at a weekly meeting or by a recorded message within 48 hours. Changes that touch an assignment brief, a due date or a marking requirement go to the tutor first. No approval is assumed from silence.

1. **Raise.** Any member records the proposal and the reason on the day it arises.
2. **Log.** The change receives an ID and the status Pending.
3. **Assess.** The package owner and other team members assess the effect on scope, schedule (including the critical path), effort, risk, quality and usability.
4. **Decide.** The group approves, rejects or defers the change, and records who decided and when.

5. **Implement.** The scope baseline, WBS dictionary, Gantt chart, critical path and risk register are updated together, and the baseline version number is increased.
6. **Verify and close.** A reviewer confirms that the updated documents agree with each other, and the status becomes Implemented.

Scope baseline v1 (Section 4 and the WBS dictionary) and schedule baseline v1 (the Gantt chart and critical path) are saved when the team approves this plan. Each approved change creates the next version, and the change log records which documents changed. Urgent changes may be discussed asynchronously, but they are still logged and reviewed at the next meeting.

9. Closure and acceptance

Assignment 1 is approved for submission by the team when the requirements and rubric map are completed, WBS codes are consistent with dictionary and Gantt, the critical path calculation is verified, the major risks each have owner/trigger/response, the three prototype screens and feedback states are linked, and a teammate reads that the explanation of usability has been signed off. These are the planned checks derived from the Week 3 section of the discussion on acceptance and handover (Li n.d.e, slides 30-33). The leader verifies the citation in Harvard style, 800 words without references per group--named filename, signed contribution form, true weekly minutes, then opens the PDF export to check figures and page breaks.

For the later integrated product: AC-01 to AC-03 above are achieved by providing documented sample records; the backend reproduces the indicator as displayed; failure and coverage states are shown; the model evaluation and known limitations are shown; and the handover is source, dataset manifest, setup steps and known limitations. A post-project review records what changed and what the team learned. The lecturer/tutor is the academic acceptance authority.

10. Prototype design and usability

The first clickable prototype uses **illustrative data only**, following the Week 1 guidance on prototyping (Li n.d.b, slides 14–18). Screen 1 offers a supported sample address, states the historical snapshot and limits, and prevents unsupported input. Screen 2 shows an activity timeline with daily counts, marks dates containing flagged transactions and provides a visible status/loading state. Screen 3 lists transactions for a chosen day and opens a detail panel with transaction ID, time, address-related amount, recent transaction count, comparative baseline and an explanation of why the indicator appeared. Empty, invalid and insufficient-history examples are included. The transaction-level score is a visual placeholder until A2 establishes a method and threshold.

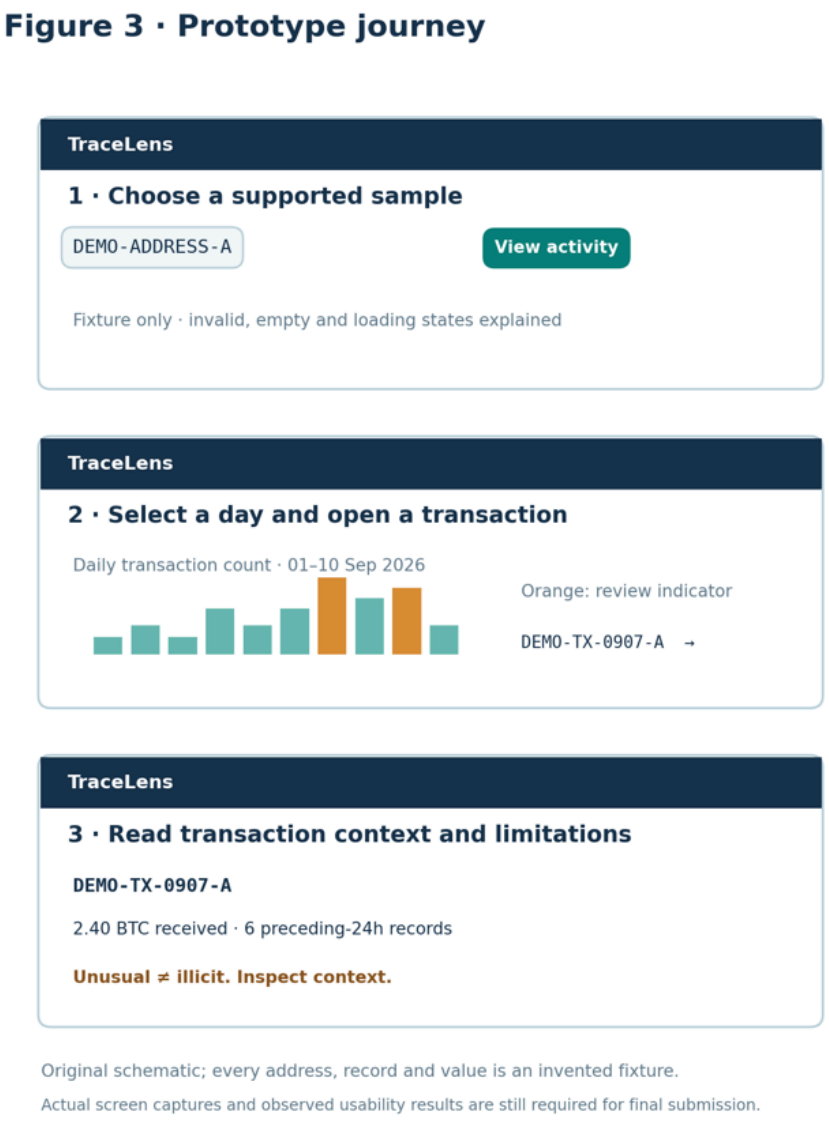


Figure 3. Original schematic of the clickable prototype; all values are illustrative.

Bitcoin_activity... pe.html
29 Sep 2026, 10:53 AM

The design uses Nielsen's heuristics (Nielsen 1994), as developed in the Week 3 usability lecture (Li n.d.e, slides 38–46 and 54–63): the loading/status message makes system status visible; “unusual activity” and units match the researcher's task; clear back/clear actions preserve control; supported examples and format guidance prevent errors; consistent labels support recognition; and the detail panel explains recovery and uncertainty in plain language. These are design claims to evaluate, not proof of usability. The UX lead asks at least two other group members or appropriate volunteer users to complete: (1) open a sample address, (2) find a highlighted date, (3) locate the transaction and explain what the indicator means, and (4) recover from an unsupported input. Record observed confusion and revisions without inventing results. The final report should include legible screenshots of the three screens, a short issue/revision table and the actual test date/participants when conducted.

11. Conclusion

The plan links the researcher's task to source feasibility, transaction-level requirements, deliverables, dependent activities, risks, acceptance checks and an interface demonstration. The source and model remain a deliberate decision gate. The team can now test one real sample, confirm the remaining Canvas date and individual owners, run a genuine usability review, and replace this draft's proposal labels with approved evidence before submission.

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